

FEA HW1

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1 Taylor and Interpolation

$$f(x) = (x + \cos(x))e^x$$

1.1

We write the second order Maclaurin series for each elementary function making up $f(x)$. x is its own second order Maclaurin series.

$$\cos(x) = 1 - \frac{x^2}{2!} + \mathcal{O}(x^3)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \mathcal{O}(x^3)$$

$$(x + \cos(x))e^x = 1 + x + \frac{x^2}{2!} - \frac{x^2}{2!} - \frac{x^3}{2!} - \frac{x^4}{2!^2} + x + x^2 + \frac{x^3}{2!} + \mathcal{O}(x^3)$$

So the first three nonzero terms and truncation error of the function are

$$f(x) = 1 + 2x + x^2 + \mathcal{O}(x^3)$$

1.2

If the desired truncation error is $\mathcal{O}(h^4)$ then the approximating polynomial must be of order 3. This means we need 4 nodal points to interpolate with Lagrange polynomials, and 4-element vectors. Since the nodal points must be equally spaced in the interval,

$$f_i = [f(0) \quad f(\frac{1}{6}) \quad f(\frac{1}{3}) \quad f(\frac{1}{2})]^T$$

$$\Rightarrow f_i = [1 \quad 1.3619 \quad 1.7840 \quad 2.2712]^T$$

$$\Rightarrow N_i = \left[\frac{(x-\frac{1}{6})(x-\frac{1}{3})(x-\frac{1}{2})}{-\frac{1}{36}} \quad \frac{x(x-\frac{1}{3})(x-\frac{1}{2})}{\frac{1}{108}} \quad \frac{x(x-\frac{1}{6})(x-\frac{1}{2})}{-\frac{1}{108}} \quad \frac{x(x-\frac{1}{6})(x-\frac{1}{3})}{\frac{1}{36}} \right]$$

1.3

See MATLAB file.

1.4 1D Integration

See MATLAB file.

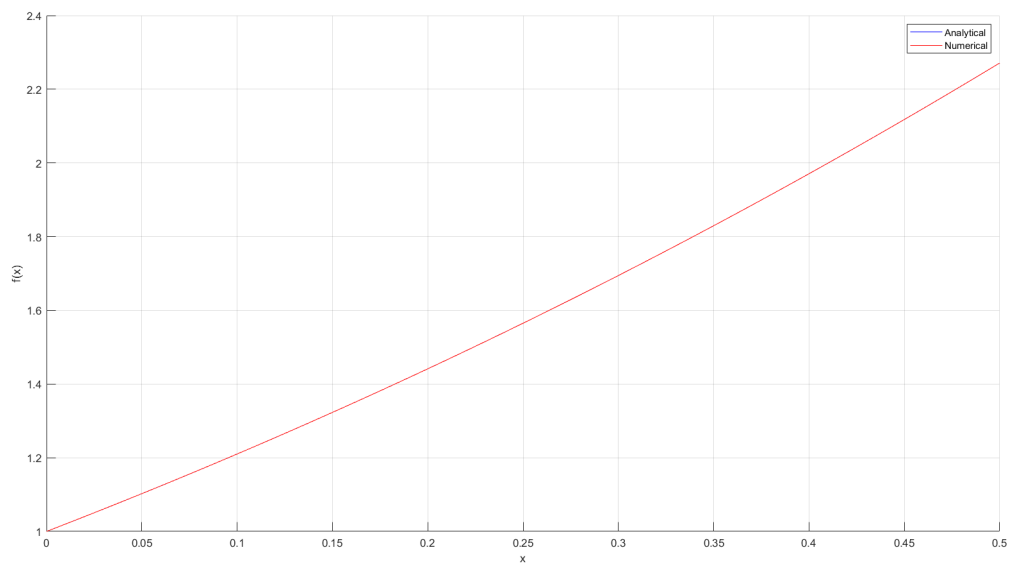


Figure 1: Q1.3: $f(x)$ vs. Second order Interpolation of $f(x)$

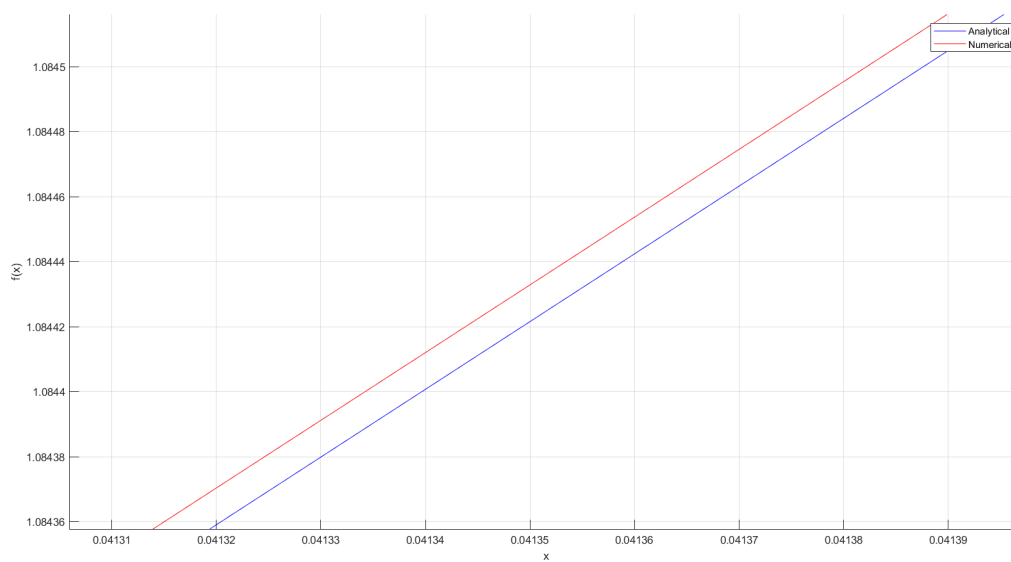


Figure 2: Q1.3: Figure 1, zoomed in to show the two curves

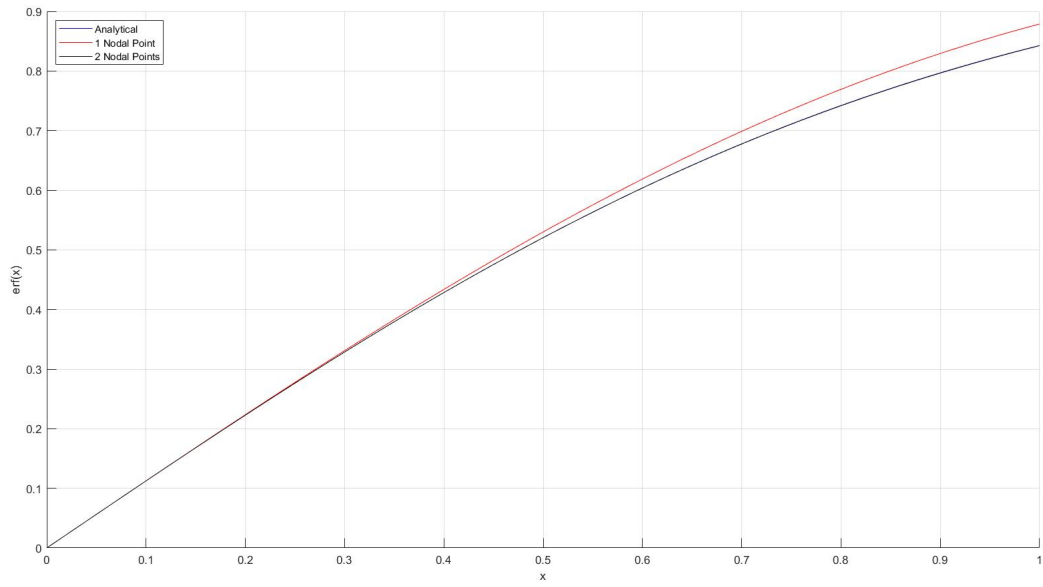


Figure 3: Q2: $\text{erf}(x)$ vs. Various Gauss Quadratures of $\text{erf}(x)$

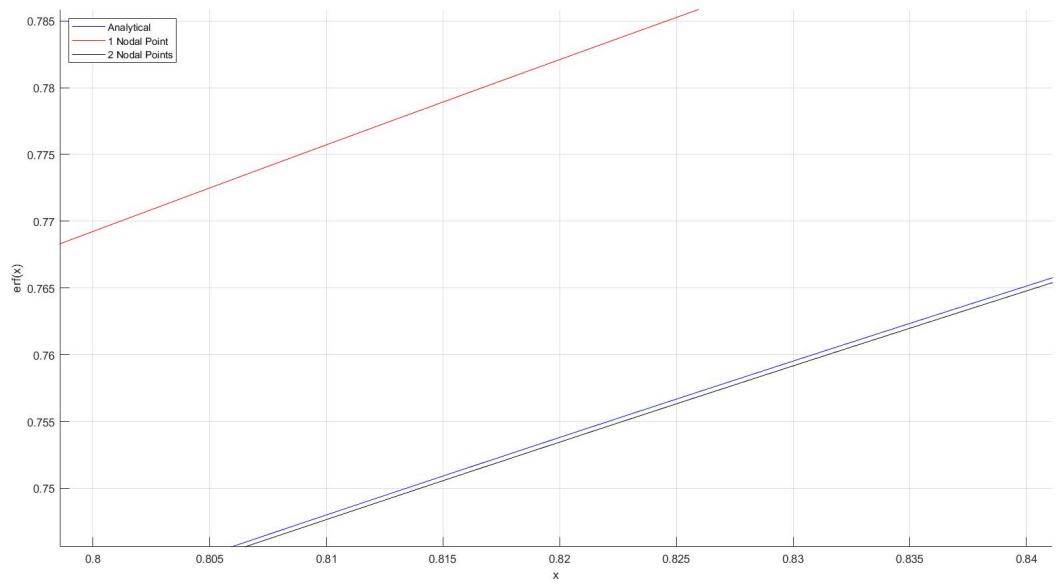


Figure 4: Q2: Figure 3, zoomed in to show the three curves